

Conditional *wh*-questions with VP Ellipsis *

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1. Introduction

This paper concerns a construction I call *conditional wh-questions with VP Ellipsis*, exemplified in (1a) and (1b). Descriptively speaking, they are root object *wh*-questions that appear with an *if*-clause, which in turn contains an elided VP. For convenience, I will refer to them simply as *conditional wh-questions*.

- (1) a. What museum would you visit if SALLY did Δ ?
Paraphrase: What museum x is such that you would visit x if Sally visited x ?
b. If you COULD Δ , what advice would you give to your younger self?
Paraphrase: If you could give advice to your younger self, what advice x is such that you would give x to your younger self?

These examples are annotated with their most natural paraphrases. We see that the interpretation of the elided object is different in these examples. (1a) has a *bound reading*, where the elided object is interpreted as a *wh*-bound variable, while (1b) has an indefinite reading, where the elided object is interpreted as an indefinite.

Conditional *wh*-questions with the indefinite reading (1b) are interesting because there is a syntactic mismatch between the elided object and its antecedent; the elided object is a non-*wh* indefinite, while its antecedent is a *wh*-phrase (or a *wh*-trace). Typically, ellipsis

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Notations in this paper: UPPERCASE denotes accented material. Δ denotes an ellipsis site. Elided material are ~~enclosed in angled brackets and struck through~~. A focus-marked constituent is indicated like _F[this]. Binding indices are subscripted on linguistic expressions that behave like variables (e.g. it_2 , t_3) and superscripted on those that behave like binders (e.g. $[every\ student]^4$, $[which\ dog]^5$). For a type τ , $\{\tau\}$ is the type of sets of objects of type τ . For types τ_1 , τ_2 , the function type $\langle\tau_1, \tau_2\rangle$ is abbreviated as $\tau_1\tau_2$. This abbreviation is right-associative: e.g., est is $\langle e, st \rangle$, not $\langle es, t \rangle$. In lambda-terms, variable binders will be subscripted with the type of the variable it binds (e.g. $\lambda x_e, \forall p_{st}, \exists f_{et}$) Metalinguistic constants are indicated with sans-serif font (e.g., dog, believe). For a type τ , D_τ is the set of semantic objects of type τ .

is thought to be licensed by an identity condition, which requires the elided constituent and its antecedent to be identical in a relevant aspect. What exactly is the identity of this condition? Examples of ellipsis like (1b) that feature an apparent mismatch between elided material and its antecedent particularly informative for our investigation, because they tell us what kinds of mismatch the identity condition tolerates. The first contribution of this paper is to document conditional *wh*-questions as an example of mismatch in VP Ellipsis.

Another question about conditional *wh*-questions is what controls the distribution of the bound and indefinite readings of the elided object. Using structural diagnostics, I show that the crucial factor is the height at which the *if*-clause is attached to the *wh*-question. Specifically, the elided object has a bound reading if the *if*-clause is inside the *wh*-question; otherwise, it has an indefinite reading. The second contribution of this paper is to argue for this height-interpretation correlation, and point out that the explanation for this correlation come from a theory of ellipsis.

This paper is structured as follows. I first show the height-interpretation correlation (Section 2), then develop an analysis of VP Ellipsis that explains this correlation (Section 3). I conclude in Section 4.

2. The height of the *if*-clause controls the interpretation of the elided object

The distribution of the two readings of the elided object is described in (2).

(2) *Height controls interpretation*

- a. An LF with a *low if*-clause, i.e., an *if*-clause attached inside the *wh*-question, can have the bound reading but not the indefinite reading.
- b. An LF with a *high if*-clause, i.e., an *if*-clause that isn't attached inside the *wh*-question,¹ can have the indefinite reading but not the bound reading.

(3) illustrates (2) with LFs. INDEF is a determiner that contributes an indefinite meaning; its exact identity is undetectable because it is elided.

- (3) a. *A low if-clause only allows a bound reading*
 [What museum]¹ would [_{VP} [_{VP} you visit *t*₁]
 [if Sally did ⟨visit { *it*₁ / *INDEF museum } ⟩]]?
- b. *A high if-clause only allows an indefinite reading*
 [[What advice]¹ would you give *t*₁ to your younger self]
 [if you could ⟨give { **it*₁ / INDEF advice } to your younger self ⟩]]?

Note that we would *a priori* expect the bound reading to be impossible in a high *if*-clause (3b) if we assume binding requires c-command. However, there is no reason to expect the

¹I avoid describing high *if*-clauses as *if*-clauses attached *outside* the *wh*-question, because there is no evidence that such *if*-clauses are attached to their *wh*-questions at all. My subsequent discussion of ellipsis doesn't require high *if*-clauses to be externally attached either; they just have to be *not inside* the *wh*-question.

indefinite reading to be impossible in a low *if*-clause (3a); the explanation for this fact has to come from a theory of ellipsis.

I now defend (2) using arguments from three structural diagnostics from the literature on the syntax/semantics of conditionals (Iatridou 1991, Bhatt and Pancheva 2006).

2.1 Condition C

A Condition C effect arises when a pronoun c-commands a coreferential name. Consider a conditional *wh*-question where the *wh*-question has a pronominal subject, and the *if*-clause has a coreferential name as its subject. By (2), we expect a Condition C effect when and only when the elided object has a bound reading. This is indeed the case: the indefinite reading is possible, with either an utterance-initial (4a) or an utterance-final *if*-clause (4b). The bound reading is impossible (4c).²

- (4) a. If Sally₂ COULD ~~⟨give advice to her₂ younger self⟩~~,
[what advice]¹ would she₂ give *t*₁ to her₂ younger self?
b. [What advice]¹ would she₂ give *t*₁ to her₂ younger self,
if Sally₂ COULD ~~⟨give advice to her₂ younger self⟩~~?
c. *[What book]¹ would he₂ read *t*₁ if John₂ was told to ~~⟨read it₁⟩~~?

2.2 Quantifier binding

Consider a conditional *wh*-question where the *wh*-question has a quantificational subject like *no woman*. Assume binding requires c-command. By (2), we expect this subject to be able to bind into the *if*-clause if and only if the elided object has the bound reading. This is indeed the case: the bound reading allows this kind of quantifier binding (5a), but the indefinite reading doesn't, no matter if the *if*-clause is final (5b) or initial (5c).

- (5) a. [What book]¹ would [no man]² read *t*₁
if his₂ parents told him₂ not to ~~⟨read it₁⟩~~?
b. *[What advice]¹ would [no woman]² give *t*₁ to her₂ younger self,
if she₂ could ~~⟨give advice to her₂ younger self⟩~~?
c. *If she₂ could ~~⟨give advice to her₂ younger self⟩~~,
[what advice]¹ would [no woman]² give *t*₁ to her₂ younger self?

2.3 If-then

The *if*-clause in constructions of the form *if P then Q* is commonly analyzed as base-generated as an adjunct to *then Q* (Iatridou 1991, Izvorski 1996, Collins 1998, Bhatt and Pancheva 2006). Conditional *wh*-questions can also take on this form: they have an initial *if*-clause and a *then* prefixed to the *wh*-question part. By (2), we expect such a conditional *wh*-question to only have an indefinite reading. This is indeed the case.

²The bound reading isn't possible with an initial *if*-clause either, though I don't show it.

- (6) a. If you could ~~give advice to your younger self~~,
 then [what advice]¹ would you give t_1 to your younger self?
 b. *If Sally did ~~visit it_T~~,
 then [what museum]¹ would you visit t_1 ?

2.4 The bound and indefinite readings deserve different LFs

A lingering concern at this point might be that maybe the bound and indefinite readings are not generated by different LFs. Perhaps conditional *wh*-questions have a unified structure at LF where the elided object is consistently interpreted with one reading, from which the other reading follows by means of entailment, inference, etc. I reject this alternative hypothesis because there are contexts that make potential answers to the two readings of the same conditional *wh*-question different.

An example of such a context is (7). The conversations (8a) and (8b) are both felicitous under (7). The conditional *wh*-questions in the two conversations form a minimal pair, only varying in the interpretation of the elided object (and the linear position of the *if*-clause, manipulated just to facilitate the two readings); yet, they induce different responses. From now on, I will assume that the two readings are generated by distinct LFs; otherwise, it is difficult to see how one can account for the presence of both readings of the elided object in (8a) and (8b).

- (7) Context: Bill is a big fan of Kraftwerk. He also doesn't like being told what to do. If you ask him to play a specific Kraftwerk song, he always plays a different song from what was requested. But if you ask him to play a song of his choice, he always plays his favorite, "Autobahn."

- (8) a. Person 1: [Which Kraftwerk song]¹ would Bill₂ play t_1
 if he₂ were asked to ~~play it_T~~?
 Person 2: No song.
 b. Person 1: If he₂ were asked to ~~play INDEF Kraftwerk song~~,
 [which Kraftwerk song]¹ would Bill₂ play t_1 ?
 Person 2: "Autobahn."

3. Ellipsis analysis

Now, I will describe an analysis of how ellipsis is licensed in conditional *wh*-questions. This analysis needs to explain two things. First, it must explain why VP Ellipsis is licensed when the elided object has an indefinite reading, despite the apparent syntactic mismatch; the elided object is an indefinite, while its antecedent is a *wh*-phrase. Second, it must explain (2).

I will work with the PF-deletion framework of ellipsis licensing theories, whereby the phonological deletion of an elided constituent *E* is licensed by its satisfaction of an identity condition with some antecedent *A*. The problem, then, is how this condition should be

formulated: does it require *E* and *A* to be identical in their syntax, or in their semantics, or both?

My first explanandum, namely the mismatch in conditional *wh*-questions with the indefinite reading like (1b), guides my response to this problem. Even though the exact form of the elided indefinite can't be detected, we may assume that there is a genuine difference between the syntactic properties of this indefinite and those of the *wh*-phrase, which presumably is the elided indefinite's counterpart in the antecedent VP. For example, a non-*wh* indefinite doesn't need to be licensed by an interrogative head, while a *wh*-phrase does; this is why an interrogative head isn't present in the *if*-clause in (1b). Thus, the licensing condition for VP Ellipsis can't be one that requires *E* and *A* to be strictly syntactically identical, because (1b) violates such a condition.

More generally, VP Ellipsis is known to tolerate certain kinds of syntactic mismatch between DPs in the elided and the antecedent VPs, an effect sometimes known as *vehicle change* (Fiengo and May 1994); some examples are given in (9). These examples, just like the indefinite reading of conditional *wh*-questions, all challenge a strict syntactic identity condition.

- (9) a. *Name vs. pronoun* (Merchant 2018:ex. 63)
 They [_A arrested Alex₃], though he₃ thought they wouldn't $\langle \textit{E-arrest-him}_3 \rangle$.
 b. *Negative polarity item vs. indefinite* (Williams 1977:ex. 123,125)
 John doesn't [_A have any horses], but Bill does $\langle \textit{E-have-some-horses} \rangle$.
 c. *Negative indefinite vs. indefinite* (Johnson 2001:ex. 103,104)
 I could [_A find no solution], but Holly might $\langle \textit{E-find-a-solution} \rangle$.

A popular reaction to these cases of syntactic mismatch is to adopt an ellipsis licensing condition that requires semantic identity. The syntactically different DPs in (9) can reasonably be construed as having equal meanings, and this very well may be the case for the indefinite readings of conditional *wh*-questions as well. After all, the semantic link between *wh*-phrases and indefinites have been widely observed (Kratzer and Shimoyama 2002); some semantic analyses even take the two to have equal denotations.

As such, I will adopt a semantic identity condition for licensing VP Ellipsis. This gives us the intuition for an answer to our first explanandum: VP Ellipsis is licensed in the indefinite reading of conditional *wh*-questions because *wh*-phrases and indefinites have the same denotation. I will now implement this intuition more formally, using alternative semantics and Heim (1997)'s theory of ellipsis licensing.

3.1 Alternative semantics

I adopt an alternative semantics, whereby the denotations of most linguistic expressions are sets. "Normal" lexical items denote singleton sets, while indefinites and *wh*-phrases denote non-trivial sets of alternatives. In this paper, I focus on indefinite DPs of the form INDEF *NP* and *wh*-phrases of the form *what NP*.

Let τ be some type. A function f of type $\langle et, \tau e \rangle$ is a τ -Skolemized choice function if $X(f(X)(y))$ for any non-empty set of entities $X \in D_{et}$ and an object $y \in D_\tau$. I write SCF_τ

for the set of τ -Skolemized choice functions. I assume that INDEF_τ and what_τ both denote SCF_τ . Some example lexical entries are given in (10).

- (10) a. $\llbracket \text{student} \rrbracket = \{\text{student}\}$ c. $\llbracket \text{INDEF}_\tau \rrbracket = \text{SCF}_\tau$
b. $\llbracket \text{read} \rrbracket = \{\text{read}\}$ d. $\llbracket \text{what}_\tau \rrbracket = \text{SCF}_\tau$

The usual syncategorematic rules are extended to work with alternative meanings; for example, Function Application becomes Pointwise Function Application (11).

(11) *Pointwise Function Application*

If A, B are trees, $\llbracket A \rrbracket$ has type $\{\tau_1 \tau_2\}$ for some types τ_1, τ_2 and $\llbracket B \rrbracket$ has type $\{\tau_1\}$, then $\llbracket A B \rrbracket$ has type $\{\tau_2\}$, defined by

$$\llbracket A B \rrbracket^g = \{f(x) \mid f \in \llbracket A \rrbracket^g, x \in \llbracket B \rrbracket^g\}.$$

Indefinites interact with $\exists$, an operator associated with the syncategorematic rule (12). Effectively, it flattens a set of propositions by existential quantification into a singleton set containing exactly one proposition. With this rule, we can derive the denotation of a sentence containing an indefinite, like (13); its LF is (13b) and its denotation is (13c).³

(12) *Existential Closure*

If A is a tree and $\llbracket A \rrbracket$ has type $\{st\}$, then $\llbracket \exists A \rrbracket$ has type $\{st\}$, defined by

$$\llbracket \exists A \rrbracket^g = \{\lambda w_s. \exists p_{st} \in \llbracket A \rrbracket^g. p(w)\}$$

- (13) a. Every student read INDEF_e paper.
b. $\exists$ [every student] $\lambda 1 t_1$ read $\llbracket \text{INDEF}_e \text{ paper} \rrbracket x_1$]
c. $\llbracket (13a) \rrbracket^g = \{\lambda w_s. \forall x_e. \text{student}(x) \rightarrow \exists f_{\langle et, ee \rangle} \in \text{SCF}_e. \text{read}(f(\text{paper}))(x))(w)\}$

Wh-phrases also interact with an operator Q , which is associated with the rule (14). It is semantically vacuous. With this rule, we can derive the denotation of a *wh*-question like (15) if we assume the *wh*-phrase is reconstructed; the LF is (15b) and the denotation is (15c), a set of possible answers (Hamblin 1973, Karttunen 1977).

(14) *Question*

If A is a tree and $\llbracket A \rrbracket$ has type $\{st\}$, then $\llbracket Q A \rrbracket$ has type $\{st\}$, defined by

$$\llbracket Q A \rrbracket^g = \llbracket A \rrbracket^g$$

- (15) a. What integer succeeds every integer?
b. Q [what integer] succeeds every integer?
c. $\llbracket (15a) \rrbracket^g = \{\lambda w_s. \forall x_e. \text{integer}(x) \rightarrow \text{succeeds}(x)(f(\text{integer}))(x))(w) \mid f \in \text{SCF}_e\}$

³In order to interpret the lambda-abstraction created by QR of *every student*, we need Predicate Abstraction, which is known to not mix well with alternative semantics. An adequate formulation of Predicate Abstraction is possible if we relax the assumption that all linguistic expressions denote sets (Shan 2004, Charlow 2020, Bumford 2022); the consequent changes to the system described here doesn't affect any of my upcoming ellipsis analysis.

3.2 Heim (1997)’s theory of ellipsis licensing

I work with a variant of Heim (1997)’s theory of ellipsis licensing. It originally comes in three parts: a VP-level syntactic identity condition, a semantic identity condition (16), and a constraint against meaningless coindexing (17). I drop the syntactic identity condition because it incorrectly rules out the indefinite reading of conditional *wh*-questions; it is also known to be too strong in other respects (Merchant 2018). The semantic identity condition (16) allows lexical differences between *E* and *A* to be neutralized by focus-marking the parts of *E* responsible for any such differences. $\llbracket E \rrbracket_f$ is the union over all $\llbracket E' \rrbracket$, where *E'* is obtained by replacing some focus-marked parts of *E* with linguistic expressions of the same semantic type. *E* and *A* are also called *parallelism domains*.

- (16) An elided VP must be contained in a constituent *E* which contrasts with a constituent *A*. We say *E* *contrasts with* *A* iff:
- a. *E* and *A* don’t overlap; and (overlap condition)
 - b. *E* and *A* have equal denotations up to any focus-marked parts in *E*, or, more formally, $\llbracket A \rrbracket \subseteq \llbracket E \rrbracket_f$. (focus condition)
- (17) *No Meaningless Coindexing*
 If an LF contains an occurrence of a variable *v* that is bound by a node α , then all occurrences of *v* in this LF must be bound by α .

3.3 The indefinite reading of conditional *wh*-questions

I now show how ellipsis is predicted to be licensed for conditional *wh*-questions with the indefinite reading. According to (2), the indefinite reading is possible only if the *if*-clause is high. That LF is (18). I assume all head and A-movement to be reconstructed.

Assuming the *wh*-phrase is reconstructed to its base position, we can locate the parallelism domains *E* and *A* as indicated.⁴ They don’t overlap, passing the overlap condition (16a). Since *what_s* and INDEF_{*s*}, where *s* is the type of possible worlds, have the same denotation, the only difference between $\llbracket E \rrbracket$ and $\llbracket A \rrbracket$ is lexical, between $F[COULD]$ and *would*. This difference is neutralizable by focus-marking; thus, the focus condition is satisfied (16b), as shown more formally in (18a) and (18b). Ellipsis is correctly predicted to be licensed. $D_{\langle st, st \rangle}$ is the set of functions from propositions to propositions; crucially, $\llbracket would \rrbracket, \llbracket could \rrbracket \in D_{\langle st, st \rangle}$. This concludes my explanation for the first explanandum.

- (18) [Q_A would you give [what_{*s*} advice] to your younger self]
 [if $\exists E F[COULD]$ you $\langle$ give INDEF_{*s*} advice to your younger self $\rangle$]?
- a. $\llbracket A \rrbracket = \{ would(\lambda w_s. give(f(advice)(w))(you)(w)) \mid f \in SCF_s \}$
 - b. $\llbracket E \rrbracket_f = \{ g(\lambda w_s. give(f(advice)(w))(you)(w)) \mid f \in SCF_s, g \in D_{\langle st, st \rangle} \}$

⁴This account works even if the *wh*-phrase isn’t *in situ*.

The second thing we need to explain is (2). To do so, we need to account for why the indefinite reading is blocked when the *if*-clause is low. Such an LF is (19), and we seem to incorrectly predict ellipsis to be licensed with the indicated parallelism domains.

- (19) Q would $[_{VP}$ $\boxed{A \text{ you give } [\text{what}_s \text{ advice}] \text{ to your younger self}}$
 $[\text{if } \exists_F[\text{COULD}] \boxed{E \text{ you } \langle \text{give-INDEX}_s \text{ advice to your younger self} \rangle}]]?$

Note that we are only able to find non-overlapping parallelism domains in (19) if we ignore the operators $\exists$ and Q , which the alternative-generating expressions in E and A associate with. For a constituent X , let's write $\lceil X \rceil$ for the smallest constituent containing X as well as all operators associated with some alternative generator in X . If the overlap condition required $\lceil E \rceil$ and $\lceil A \rceil$ to be non-overlapping, then it would be impossible to find appropriate E and A in our LF, as shown in (20) (PP abbreviates *to your younger self*). The low-*if*-clause case exhibits inescapable antecedent containment, which bans VP Ellipsis.

- (20) Q would $[_{VP} [\text{you give } [\text{what}_s \text{ advice}] \text{ PP }]$
 $\boxed{\lceil A \rceil \quad [\text{if } \exists \boxed{\lceil E \rceil_F [\text{COULD}] \text{ you } \langle \text{give-INDEX}_s \text{ advice PP} \rangle }]] ?$

However, the focus condition can't require $\lceil E \rceil$ and $\lceil A \rceil$ to be semantically equal; otherwise, we would lose the account for the high-*if*-clause case (18) since the operators $\exists$ and Q make different semantic contributions. I thus propose a new semantic identity condition (21); only the overlap condition is changed. This correctly accounts for why the indefinite reading requires a high *if*-clause.

- (21) An elided VP must be contained in a constituent E which contrasts with a constituent A . We say E *contrasts with* A iff:
- a. $\lceil E \rceil$ and $\lceil A \rceil$ don't overlap; and (new overlap condition)
 - b. E and A have equal denotations up to any focus-marked parts in E , or, more formally, $\llbracket A \rrbracket \subseteq \llbracket E \rrbracket_f$. (focus condition)

3.4 The bound reading of conditional *wh*-questions

To complete the explanation for (2), we need to account for why the bound reading requires a low *if*-clause. The bound reading belongs to a class of VP Ellipsis configurations known as *Co-binding*, where a common binder binds both the elided and the antecedent objects at LF. I adopt Takahashi and Fox (2005)'s analysis of Co-binding examples to account for the bound reading, which I briefly describe below.

The LF for the bound reading with a low *if*-clause is illustrated in (22). The *wh*-phrase moves to semantically bind the elided object, and is interpreted *ex situ*. I assume it is interpreted at [Spec; ν P] as indicated in (22) rather than [Spec; CP], as this makes the upcoming analysis compatible with accounts of crossover effects that bans pronominal binding from

A'-positions (Reinhart 1983, van Riemsdijk and Williams 1981).⁵ Further, I assume that the elided object is a pronoun (Kennedy 1997).

- (22) Q would [what museum] $\lambda 1$ [_{VP} A you visit t_1] [if E $F[SALLY]$ did $\langle \text{visit-it}_T \rangle$]]?
- a. $\llbracket A \rrbracket^g = \{\text{visit}(g(1))(\text{you})\}$
b. $\llbracket E \rrbracket_f^g = \{\text{visit}(g(1))(x) \mid \text{person}(x)\}$

In this case, the parallelism domains as indicated are non-overlapping and satisfy the focus condition: their denotations are only different by the outermost argument of visit, which is neutralized by focus-marking on $F[SALLY]$ in E . The new overlap condition (21a) doesn't check beyond E and A because $\llbracket E \rrbracket = E$ and $\llbracket A \rrbracket = A$, as neither E nor A contains any alternative generators. Ellipsis is correctly predicted to be licensed.

Assume binding requires c-command. Then we *a priori* rule out a bound reading with a high *if*-clause, because binding is impossible. But what about (23), a LF with a low *if*-clause where the elided and antecedent objects are coindexed? The elided object is bound by the *wh*-phrase, but the antecedent object is interpreted as a free variable, referring to a contextually salient entity. This interpretation is empirically impossible, but our theory seems to predict it to be grammatical, with the indicated parallelism domains. However, this LF violates (17) as it contains an instance of “meaningless coindexing:” there are two occurrences of variable with index 1, but they are not bound by the same binder. Thus (17) correctly rules this LF out. We have correctly accounted for why the bound reading requires a low *if*-clause, thus fully explaining (2).

- (23) [Q would [what museum] $\lambda 1$ [_{VP} A you visit t_1]]
[if E $F[SALLY]$ did $\langle \text{visit-it}_T \rangle$]]?

4. Conclusion

In a conditional *wh*-question, the object of elided VP may receive either a *wh*-bound or an indefinite interpretation. I propose an analysis for this construction that accounts for two key facts: that VP Ellipsis allows a *wh*-phrase to antecede the ellipsis of an indefinite DP, and that the height at which the *if*-clause is attached to the *wh*-question determines the interpretation of the elided object. In future work, I'd like to explore ways to derive the ellipsis licensing condition I propose (21) without stipulating it.

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⁵I thank Gary Thoms for suggesting this assumption.

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